

A Critical Introduction to Scientific Realism

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constrain our epistemological investigations – which is to work within a very tight intellectual circle indeed. The chapter concludes with a summary of the key points discussed (§3.4) and some suggestions for further reading (§3.5).

3.1. Cosmic coincidences

An early articulation of this new realist intuition was offered by J. J. C. Smart (1963), who presented what he called a *plausibility argument* for scientific realism. The basic idea is that even if we could somehow dispense with our theoretical vocabulary through some process of explicit definition or instrumentalist elimination, our resulting picture of science would leave a great many things unexplained. For the eliminative instrumentalist, for example, our scientific theories are to be understood as tools for the systematization of our observational experience – they are purely symbolic devices that state that whenever a particular observational state of affairs obtains, another particular observational state of affairs will follow. But there is nothing in the eliminative instrumentalist's picture that can tell us *why* these regularities hold. On this account, it is simply a 'cosmic coincidence' that our observational experiences are related in this way, and regardless of the technical merits of the position, this is just unacceptably mysterious. Smart writes:

Is it not odd that the phenomena of the world should be such as to make a purely instrumental theory true? On the other hand, if we interpret a theory in the realist way, then we have no need for such a cosmic coincidence: it is not surprising that galvanometers and cloud chambers behave in the sort of way they do, for if there are really electrons, etc., this is just what we should expect. (1963: 39)

Suppose that we are attempting to explain why it is that different magnetic fields produce different patterns in a saturated cloud chamber. For the scientific realist, there is an extremely straightforward story that can be told. It tells us that radioactive materials emit subatomic particles which ionize the gas in the cloud chamber as they pass, leaving a visible trail in their wake; moreover, many of these subatomic particles are intrinsically charged, and will therefore be attracted or deflected in the presence of a magnetic field, resulting in a corresponding change in their path taken through the cloud chamber. Such an account clearly involves commitment to various unobservable entities and processes, but it is nevertheless an integrated and intelligible account of the phenomena we are trying to explain. By contrast, the story provided by the eliminative instrumentalist seems to depend more upon dumb luck than any

kind of causal mechanism. On this account, the most that we can say is that it is a brute fact of nature that whenever certain materials – the ones that glow in the dark and make you ill – are placed near a cloud chamber, various tracks appear; moreover, it is also a brute fact of nature that placing magnets in the vicinity of such an experimental set-up makes some of these tracks turn into spirals. There is, however, no particular reason to suppose that these facts are related.

The idea then is that scientific realism is just intrinsically more plausible than any of its instrumentalist rivals. We should note, however, that this isn't merely the old dispute as to whether or not our scientific theories should both systematize *and* explain our observational experiences, and nor is Smart simply stating that our scientific theories can provide us with knowledge of the unobservable world – an accomplishment that the eliminative instrumentalist would in any case reject as irrelevant to our scientific practice. Rather, the claim is that scientific realism provides us with a better account of our *observational experiences* and why we should expect the world to turn out the way that our theories say that it will. This is a goal that both realists and instrumentalist share, and therefore promises to provide a positive argument for scientific realism.

Nevertheless, at the centre of Smart's argument lies the idea of one philosophical position just being intrinsically more plausible than another. One can certainly feel the force of his reasoning, but at the end of the day the strength of the argument relies upon our pre-theoretical intuitions about what makes one account better than another. It therefore follows that if another philosopher simply denied that they found scientific realism more plausible than its rivals, there would be very little that Smart could say to try and convince them otherwise. For example, a committed instrumentalist might well maintain that while scientific realism offers a more intellectually pleasing account of the relationship between our observational experiences, none of this assists us in the fundamental task of deductive systematization, and is therefore actually superfluous to the real issues relating to the scientific realism debate. Or to take a more contemporary example, an empiricist like Bas van Fraassen might well maintain that while the instrumentalist explanation for the tracks in the cloud chamber leaves a lot unexplained, the scientific realist is really in no better a situation – it is not a virtue of an explanation to merely replace one mystery with another, and while talk about electrons and charge may explain the tracks in the cloud chamber, we are simply left with another set of unanswered questions about why electrons have such-and-such properties and act in the way that they do (we will return to some of these issues below). The point then is not that Smart is wrong about the plausibility of scientific realism, but only that the notion of plausibility under discussion seems to be too subjective to help us make any progress in the debate.

3.1.1. A scientific explanation for scientific success

The more contemporary articulation of this general realist intuition is usually credited to Hilary Putnam (1975a) and Richard Boyd (1984). There are a number of ways in which to understand how this version of the argument differs from – and thus attempts to improve upon – the account given by Smart. In both cases, the idea is to show how scientific realism is somehow more plausible than its rivals and can therefore provide a better explanation for why our scientific theories manage to be predictively successful. In the case of Smart, however, this notion of plausibility was to be understood in terms of our *a priori* intuitions and was therefore powerless to convince those who simply did not share these assumptions. In the case of Putnam and Boyd, by contrast, we are presented with an *a posteriori* argument for scientific realism, one where the central notion of plausibility is to be grounded in our everyday inferential practices.

More specifically, the idea is that since there is no way to adjudicate between competing *a priori* standards of plausibility, we should instead appeal to those standards of plausibility already in use within our most successful empirical investigations. Both realists and instrumentalists of course agree that our scientific theories are predictively successful; it is, after all, willingness to endorse this epistemological starting point that defines the scientific realism debate. Presumably then, both realists and instrumentalists will agree that whatever standards and criteria we use to adjudicate between competing scientific theories are themselves generally reliable, since they have allowed us to select those which are predictively successful. The proposal then is that we can appeal to these same standards and criteria to adjudicate between competing philosophical theories about the status of our scientific theories. The contemporary articulation of the realist intuition is therefore an explicitly *scientific argument*, the thought being that by looking at our most successful scientific investigations, we can adopt standards of plausibility that all sides to the scientific realism debate can accept. As Putnam puts it:

The positive argument for realism is that it is the only philosophy that does not make the success of science a miracle. That terms in mature scientific theories typically refer ... that the theories accepted in a mature science are typically approximately true, that the same terms can refer to the same even when they occur in different theories – these statements are viewed not as necessary truths but as part of the only scientific explanation of the success of science, and hence as part of any adequate description of science and its relations to its objects. (1975a: 73)

This has become the canonical articulation of our starting intuition and is known in the literature as the No-Miracles Argument for scientific realism.

In order to properly assess this reasoning, it will be helpful to outline the structure of the argument in more detail. We begin with the claim that since the approximate truth of our scientific theories is the best explanation for their predictive success, we should conclude that our scientific theories are indeed approximately true. The argument is therefore an instance of inference to the best explanation (IBE), where the idea of one explanation being better than another is to be understood in terms of those theoretical virtues by which we choose one scientific theory over another, such as simplicity, scope, coherence with the rest of our theoretical background, and so on and so forth. This gives us a more concrete sense in which one philosophical theory can be more plausible than another, but it raises the obvious question as to why we would suppose that IBE is a reliable method of inference. The idea, however, is that IBE is part of our scientific methodology, and thus once we accept that we should appeal to our scientific methodology in order to help us adjudicate between competing philosophical theories, we *ipso facto* have reason to suppose that IBE is in fact reliable. The real force behind the No-Miracles Argument is therefore a kind of epistemological slippery slope: that once we endorse a little bit of our scientific methodology in order to break our philosophical deadlock, we end up endorsing so much of that scientific methodology that we cannot help but accept scientific realism.

More specifically, we can think of the argument as proceeding in two steps. The first step begins with the claim that our scientific theories are predictively successful, something that all parties to the scientific realism debate will accept. We then note that our scientific theories do not exist in a vacuum, but are themselves the result of a number of important considerations, including for example the various constraints imposed upon our reasoning by those background theories that we already accept. To take a very simple example, if our background theories entail that a particular sort of entity cannot exist, then any contemporary scientific theory that posits the existence of such an entity will be rendered extremely unlikely, no matter what other theoretical virtues it may possess. We conclude therefore that since our contemporary scientific theories depend so heavily upon our already existing background theories, they wouldn't have turned out to be so predictively successful unless these background theories were themselves approximately true. Or alternatively, if our background theories were not themselves approximately true, then it is hard to see how the contemporary scientific theories that we select on their basis could turn out to be as predictively successful as they are.

This might seem like a somewhat roundabout way to proceed, but it is by taking this detour through our background theories that the scientific realist

is able to establish the reliability of the *a posteriori* standards of plausibility upon which his argument depends. The second step of the argument then continues with the claim that our background theories are themselves the result of previous instances of IBE. After all, we come to accept one scientific theory over another because it provides the *best explanation* for the phenomena in question, where the idea of one explanation being better than another is again to be understood in terms of the familiar theoretical virtues of simplicity, scope, coherence with other scientific theories, and so on. This claim is also fairly uncontroversial, since both realists and instrumentalists will agree that this offers an accurate description of actual scientific practice. The point though is that since we have already established in the previous step that our background theories are approximately true, the fact that they were formulated on the basis of IBE now gives us a good reason to suppose that IBE is a reliable method of inference. The rest of the argument then proceeds swiftly. According to the scientific realist, the *best explanation* for the predictive success of our scientific theories is that they are approximately true; and since we have concluded that IBE is a generally reliable method of inference, we can therefore conclude that our contemporary scientific theories are indeed approximately true as the scientific realist maintains.

3.1.2. Vicious circles and virtuous circles

Although there is certainly something compelling about the way in which the No-Miracles Argument attempts to show how even a limited endorsement of our scientific methodology inevitably leads to full-blown scientific realism, there are also a number of reasons why we might feel suspicious of this particular line of reasoning. The argument attempts to show that scientific realism offers the most scientific explanation for the predictive success of our scientific theories, and while it might well have greater simplicity and scope than its instrumentalist alternatives, there is still a sense in which it offers a very poor explanation as judged by contemporary scientific standards. The problem is that a good scientific explanation is usually one that generates novel predictions. It does not simply tell us how things have been in the past, but also makes concrete claims about how things will turn out in the future. Indeed, we saw in the previous chapter that one of the principal arguments against eliminative instrumentalism was its inability to accommodate this feature of scientific practice. But the claim that our scientific theories are approximately true does not itself lead to any novel predictions over and above those already made by the scientific theories in question (see for example Frost-Arnold 2010). This is because scientific realism is not so much a scientific explanation of predictive success as an *evaluation* of that

4.1.2. *The bankruptcy of science*

In the end then, even the most virulent form of underdetermination has very little to show us about the status of our scientific theories. The main problem is that the argument operates at too great a degree of abstraction. For while it is true that with enough ingenuity one can cook up an unlimited number of possible rivals to any predictively successful scientific theory, all equally capable of satisfying any number of additional epistemological desiderata, there will always be *some* difference between them all that the scientific realist can capitalize upon as the most reliable indicator of approximate truth. A more powerful objection to the sort of reasoning employed by the No-Miracles Argument therefore must look for more concrete considerations. Rather than showing how such an argument would fare in a number of purely hypothetical scenarios, we should instead look at the actual history of our scientific practice for instances that might undermine the central reasoning employed by the scientific realist.

One promising line of thought therefore is to note that whether or not we have good reason to believe that our current scientific theories are approximately true, one thing of which we can be certain is that they have *superseded* a number of earlier scientific theories that we now believe to be false. In fact, once we consider the issue in more detail, the history of science appears to be littered with such failures. This then raises the worry that maybe our scientific reasoning is not as reliable as we previously thought. As Henri Poincaré famously put it:

The man of the world is struck to see how ephemeral scientific theories are. After some years of prosperity, he sees them successively abandoned; he sees ruins accumulated on ruins; he predicts that the theories in vogue today will in a short time succumb in their turn, and he concludes that they are absolutely in vain. This is what he calls the bankruptcy of science. (1902 [1905]: 160)

The scientific realist argues that because our current scientific theories are predictively successful, we have good reason to suppose that they are approximately true. Yet that is precisely what earlier scientists thought, often on the basis of equally impressive predictive success. But if all of these thinkers have been proven wrong, what reasons do we have to suppose that our epistemological situation is going to be any different?

Poincaré himself did not draw any such sceptical conclusion, arguing instead that the historical record merely shows us something about the structure of our scientific theories and the limits of our scientific investigations

(this view is discussed in more detail in Chapter 6). However, many other philosophers have indeed maintained that the history of science does in fact give us good reason to reject scientific realism. This line of thought is usually referred to as the Pessimistic Meta-Induction, but we must be careful here in distinguishing between two different arguments that we might offer on the basis of our historical data. The first seeks merely to show that the positive arguments offered by the scientific realist for the approximate truth of our scientific theories are undermined by the various counter-examples produced by the track-record of our scientific theorizing. Such an argument is neither particularly pessimistic nor is it a meta-induction in any technical sense of the term. The second argument seeks to show that not only do we not have good reason to suppose our current scientific theories to be true, but that the history of science actually furnishes us with good reasons to suppose that they are in fact false. This is, of course, a much stronger conclusion to draw, and is more accurately thought of as a meta-induction, since it attempts to draw an (extremely negative) inductive conclusion *about* the inductive practices that constitute our scientific practice.

There is plenty of room for terminological confusion here, for while it is only the second argument that is accurately described by the name, it is usually the first and more modest argument that philosophers actually mean when they talk about the Pessimistic Meta-Induction (for a recent discussion of different interpretations of the argument, see Mizrahi 2013). This is because the stronger argument is simply incoherent. At a first pass, we should naturally be suspicious of any line of reasoning that attempts to provide an inductive argument as to why we shouldn't trust induction. More specifically, the argument is that since all of our past scientific theories have been shown to be false, our current scientific theories will probably be shown to be false too. The problem, however, is that the only reason we have for believing our past scientific theories to be false is because they disagree with our current scientific theories. We no longer believe in phlogiston because we now believe in the oxidization of metals; the electromagnetic field has replaced the luminiferous ether; and absolute space has succumbed to the fundamental relativity of distinct frames of reference. But if we only believe that our previous scientific theories are false on the basis of our belief that our contemporary scientific theories are true, we cannot *also* use our belief that our previous scientific theories are false to undermine our belief that our contemporary scientific theories are true.

Lipton (2000) provides a helpful illumination of this point. Suppose that we have two competing scientific theories, and perform some crucial test that provides positive evidence in favour of one and simultaneously provides negative evidence against the other. Since the first theory has been confirmed, we have reason to believe that it is true. Since the second theory has been

falsified, we have reason to believe that it is false. But then, according to the stronger reading of the Pessimistic Meta-Induction, the falsification of the second theory also tells against the first – after all, the majority of scientific theories have been falsified, and so chances are that this one will be falsified as well. But then the same experiment would provide both direct evidence in favour of a theory and indirect evidence against it. In Lipton's memorable phrase, the argument is a piece of 'judo epistemology' since it attempts to use the success of science against itself (ibid.: 199–200).

The Pessimistic Meta-Induction is therefore to be understood as the more modest proposal that the history of science offers us good reason to reject the positive arguments put forward by the scientific realist. The classic statement of this sort of argument is due to Laudan (1981), who sees the history of science as explicitly undermining the sort of reasoning exemplified by the No-Miracles Argument (similar readings of the argument are given by Psillos 1999: 101–14, and Saatsi 2005). Suppose for the sake of argument that the predictive success of a scientific theory does generally indicate the approximate truth of that theory. It would follow then that our current scientific theories are indeed approximately true as the scientific realist maintains. But our current scientific theories disagree with our earlier scientific theories, positing different types of unobservable entities and different types of causal relationships holding between them. So if our current scientific theories are approximately true, then our previous scientific theories are predominantly false. But each one of these scientific theories was predictively successful at the time. The conclusion is not then that our current scientific theories must be false, but simply that we cannot reasonably maintain that predictive success is a reliable indicator of approximate truth, despite the scientific realist's protestations to the contrary.

This might seem like something of an anticlimax. The point, however, is that rather than simply denying that predictive success is a reliable indicator of truth, or showing how predictive success is insufficiently discriminatory in a range of hypothetical scenarios, the Pessimistic Meta-Induction attempts to show that the link between predictive success and truth is in fact undermined by the scientific realist's own standards. It is therefore a serious challenge for any contemporary defence of scientific realism.

4.2. The uses and abuses of the history of science

It is difficult to deny that many of our current scientific theories offer conflicting accounts of the unobservable structure of the world than that proposed by